

Worksheet 1: August 28th

Problem 1. Let n and m be integers. We say that n *divides* m , written $n \mid m$, if there exists some other integer k such that $m = kn$.

Let a, b, c be integers. Suppose $a \mid b$ and $a \mid c$. Prove that for every other two integers m, n ,

$$a \mid (mb + nc).$$

Problem 2. Prove that if n^2 is even, then n is even.

Problem 3. We say that a positive integer n is prime if the only positive integers which divide it are 1 and n . Prove by induction that every positive integer can be written as the product of primes.

Problem 4. Let F_n denote the n -th Fibonacci number which is defined recursively via

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2}.$$

Show by induction that $F_n < 2^n$.

Problem 5. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions.

(a) Prove that if $g \circ f$ is injective, then f is injective.

(b) Is the converse true? If it is, prove it. If not, provide a counterexample.